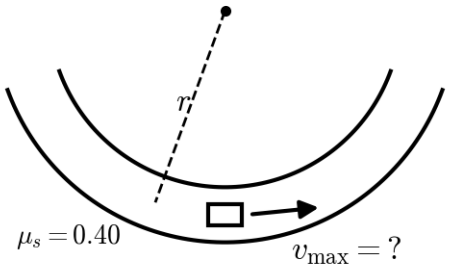
PHYS1408 Practice Exam 2-2

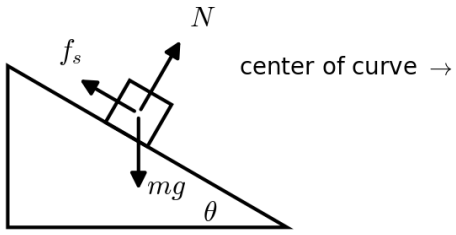
Chapters 6, 7/8, 9, 10, 11

Use $g = 9.8$ m/s^2. Choose the single best answer. Answers are rounded to the precision shown. For a minimum or maximum, choose the answer choice that actually works.

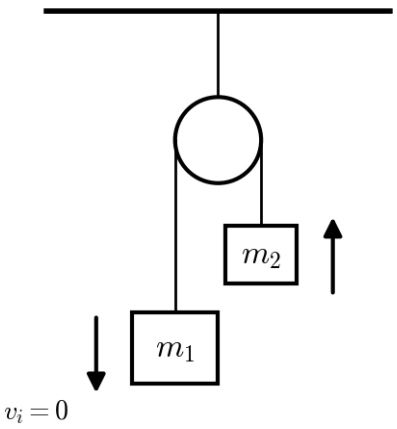
1. A car goes around a flat (unbanked) curve of radius 35 m. The coefficient of static friction between the tires and the road is $\mu_s = 0.40$. What is the maximum speed at which the car can take the curve without sliding?
a) 3.7 m/s b) 11.7 m/s c) 16.6 m/s d) 18.5 m/s
e) 137 m/s



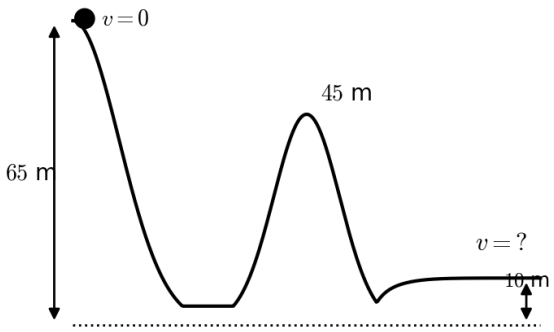
2. A highway curve of radius 90 m is banked at $\theta = 30°$. The coefficient of static friction between the tires and the road is $\mu_s = 0.15$. What is the minimum speed at which a car can take the curve without sliding down the bank?
a) 12 m/s b) 18 m/s c) 20 m/s d) 23 m/s e) 27 m/s



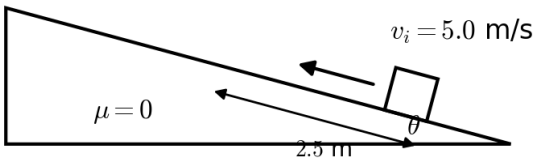
3. Two masses, $m_1 = 8.0$ kg and $m_2 = 6.0$ kg, hang from opposite ends of a light string over a massless, frictionless pulley (Atwood's machine). They are released from rest. How fast is m_1 moving after it has fallen 1.5 m?
a) 1.45 m/s b) 2.05 m/s c) 2.71 m/s d) 3.13 m/s
e) 5.42 m/s



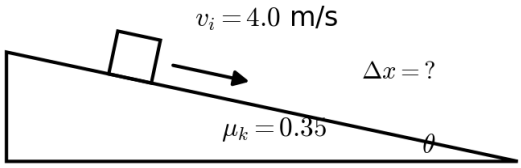
4. A roller coaster car starts from rest at the top of a 65 m hill. It goes over a 45 m hump and then down to a section of track 10 m above the ground. The track is frictionless. How fast is the car moving on the 10 m section?
a) 14.0 m/s b) 19.8 m/s c) 23.2 m/s d) 32.8 m/s
e) 35.7 m/s



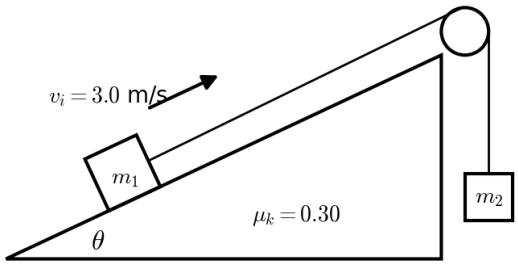
5. A block slides up a frictionless incline ($\theta = 15^\circ$) with an initial speed of 5.0 m/s. How fast is it moving after it has slid 2.5 m along the incline?
a) 3.5 m/s b) 4.3 m/s c) 6.1 m/s d) 12.3 m/s e) it stops before it slides the whole 2.5 m



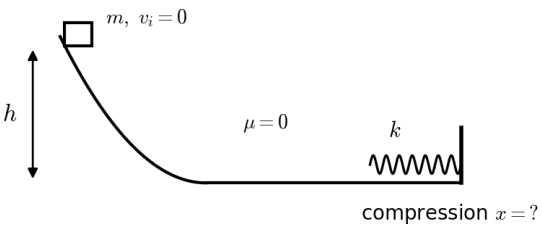
6. A block is sliding **down** a rough incline ($\theta = 12^\circ$, $\mu_k = 0.35$) at 4.0 m/s. How far does it slide along the incline before it comes to rest?
a) 1.5 m b) 2.4 m c) 6.1 m d) 12.1 m e) it never stops; it speeds up



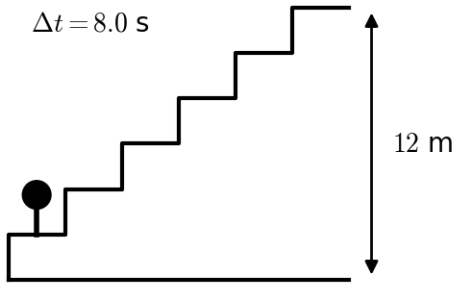
7. A 15 kg block on a 25° incline ($\mu_k = 0.30$) is tied by a light string over a massless pulley to a 4.0 kg hanging mass. The block is moving **up** the incline at 3.0 m/s (and the hanging mass is moving down). How fast is the block moving after it has slid 1.5 m up the incline?
a) 2.32 m/s b) 2.51 m/s c) 2.98 m/s d) 4.35 m/s
e) it stops before it slides the whole 1.5 m



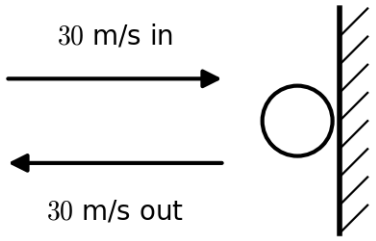
8. A 2.5 kg mass starts from rest at a height of 0.80 m on a frictionless ramp. It slides down, across a frictionless floor, and into a spring with spring constant $k = 1800$ N/m. How far is the spring compressed when the mass comes to rest?
a) 0.011 m b) 0.022 m c) 0.10 m d) 0.15 m e) 0.21 m



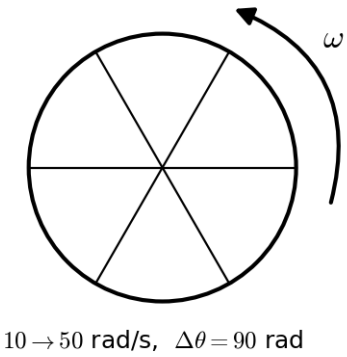
9. A 70 kg person climbs a staircase 12 m high in 8.0 s. What is the person's average power output?
a) 105 W b) 457 W c) 1030 W d) 8230 W
e) 65,900 W



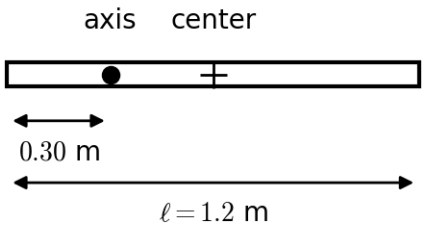
10. A 0.15 kg ball hits a wall head-on at 30 m/s and bounces straight back at 30 m/s. The ball is in contact with the wall for 0.010 s. What is the magnitude of the average force the wall exerts on the ball?
a) 0 N b) 4.5 N c) 9.0 N d) 450 N e) 900 N



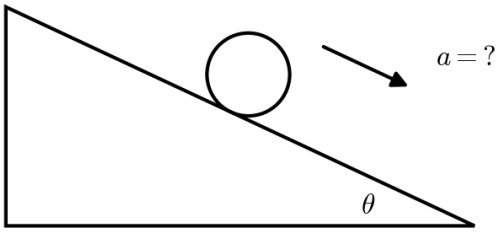
11. A wheel speeds up uniformly from 10 rad/s to 50 rad/s while turning through 90 rad. How long does this take?
a) 1.5 s b) 1.8 s c) 2.25 s d) 3.0 s e) 9.0 s



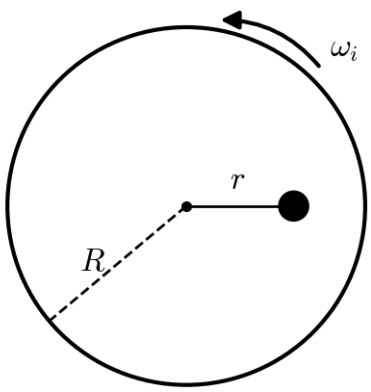
12. A thin uniform rod has mass 3.0 kg and length 1.2 m ($I_{cm} = \frac{1}{12}m\ell^2$). It rotates about an axis perpendicular to the rod, located 0.30 m from one end. What is the rod's moment of inertia about this axis?
a) 0.09 kg · m² b) 0.27 kg · m² c) 0.36 kg · m²
d) 0.50 kg · m² e) 0.63 kg · m²



13. A solid sphere ($I = \frac{2}{5}MR^2$) rolls without slipping down a 25° incline. What is the acceleration of its center down the incline?
a) 2.07 m/s² b) 2.96 m/s² c) 4.14 m/s² d) 6.34 m/s²
e) 7.00 m/s²



14. A merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 300$ kg, $R = 2.5$ m) rotates at 1.2 rad/s. A 60 kg person standing still on the ground steps straight down onto it at a distance of 1.5 m from the center. What is the new angular velocity? (Treat the person as a point mass.)
a) 1.05 rad/s b) 1.10 rad/s c) 1.12 rad/s d) 1.20 rad/s
e) 1.37 rad/s



15. A merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 400$ kg, $R = 3.0$ m) rotates at 1.5 rad/s. A uniform 60 kg rod, 3.0 m long, is dropped onto it so that it lies along a radius from the center to the edge. What is the new angular velocity?
a) 1.36 rad/s b) 1.43 rad/s c) 1.46 rad/s d) 1.50 rad/s
e) 1.73 rad/s

